State-Space Models

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Why State-Space Models?

- Many macro quantities are unobserved: output gap, latent trends, time-varying parameters.
- State-space models link observed data to hidden states that evolve over time.
- Unified framework for:
  - Trend–cycle decompositions
  - Dynamic factors
  - Time-varying parameters
  - Missing data handling

The State-Space Representation of a Dynamic System

- The vector of variables $\boldsymbol{y}_t$ observed at date t can described in terms of a possibly unobserved state vector $\boldsymbol{\alpha}_t$
- The state-space representation of $\boldsymbol{y}$ is then given by

$$\underset{n\times 1}{\boldsymbol{y}_t} = \underset{n\times k}{\boldsymbol{A}}\underset{k\times 1}{\boldsymbol{x}_t} + \underset{n\times r}{\boldsymbol{H}}\underset{r\times 1}{\boldsymbol{\alpha_t}} + \underset{n\times 1}{\boldsymbol{w}_t} \quad \text{observation equation} \quad (1)$$

$$\underset{r\times 1}{\boldsymbol{\alpha_t}} = \underset{r\times m}{\boldsymbol{B}}\underset{m\times 1}{\boldsymbol{z}_t} + \underset{r\times r}{\boldsymbol{F}}\underset{r\times 1}{\boldsymbol{\alpha_{t-1}}} + \underset{r\times 1}{\boldsymbol{v}_t} \quad \text{state equation} \quad (2)$$

$\boldsymbol{A}$, $\boldsymbol{H}$, $\boldsymbol{B}$ and $\boldsymbol{F}$ are matrices of parameters, $\boldsymbol{x}_t$ and $\boldsymbol{z}_t$ are vectors of exogenous variables, $E(\boldsymbol{w}_t\boldsymbol{w}_t') = \boldsymbol{R}$, $E(\boldsymbol{v}_t\boldsymbol{v}_t') = \boldsymbol{Q}$ and $E(\boldsymbol{v}_t\boldsymbol{w}_t') = \boldsymbol{0}$, where $\boldsymbol{Q}$ and $\boldsymbol{R}$ are $r \times r$ and $n \times n$ matrices

Example: AR(p) as a State-Space Model

- AR(p): $y_t = c + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \varepsilon_t$.
- Choose state $\alpha_t = [\, y_t,\, y_{t-1},\, \ldots,\, y_{t-p+1}\,]'$.
- Observation: $y_t = [1\ 0\ \cdots\ 0]\,\alpha_t$.
- State:

$$\alpha_t = \begin{bmatrix} \phi_1 & \phi_2 & \cdots & \phi_{p-1} & \phi_p \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{bmatrix} \alpha_{t-1} + \begin{bmatrix} \varepsilon_t \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \varepsilon_t \sim N(0, \sigma^2).$$

Example: Unobserved Components (Trend + Cycle)

- Decompose y_t into trend τ_t and cycle c_t:

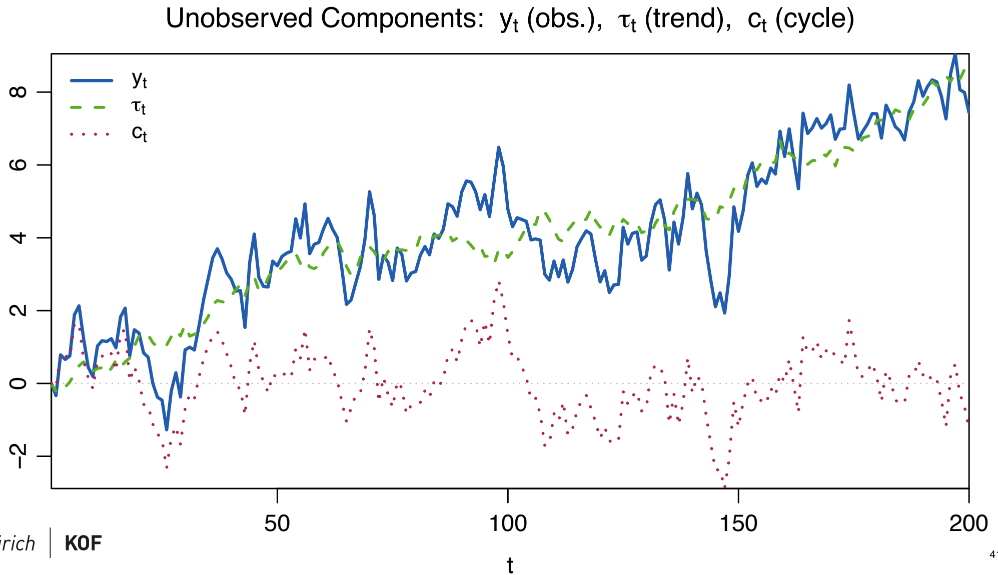
$$y_t = \tau_t + c_t, \qquad \tau_t = \delta + \tau_{t-1} + \nu_t, \qquad c_t = \phi c_{t-1} + \eta_t,$$

with $|\phi| < 1$, $\begin{bmatrix} \nu_t \\ \eta_t \end{bmatrix} \sim N(0, \Sigma)$.

- State: $\alpha_t = [\,\tau_t,\ c_t\,]'$; observation: $y_t = [1\ 1]\alpha_t$.
- State transition:

$$\alpha_t = \begin{bmatrix} 1 & 0 \\ 0 & \phi \end{bmatrix} \alpha_{t-1} + \begin{bmatrix} \delta \\ 0 \end{bmatrix} + \begin{bmatrix} \nu_t \\ \eta_t \end{bmatrix}.$$

Example: UC with Simulated Data



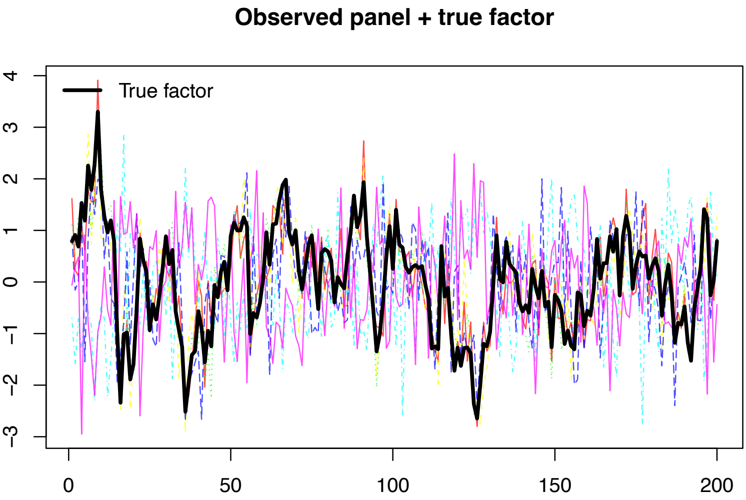
Example: Dynamic Factor Model (Activity Index)

- One common factor f_t drives n observables:

$$y_{it} = c_i + \lambda_i f_t + \nu_{it}, \quad i = 1, \ldots, n; \;\; f_t = \phi f_{t-1} + \eta_t.$$

- Observation: $y_t = A + H\alpha_t + w_t$ with $A = [c_i]$, $H = [\lambda_i]$, $\alpha_t = f_t$.
- Idiosyncratic: $w_t \sim N(0, R)$ with $R = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_n^2)$.
- Identification: fix sign/scale (e.g., $\lambda_1 > 0$, $\sigma_f^2 = 1$) or triangular H for $r > 1$.

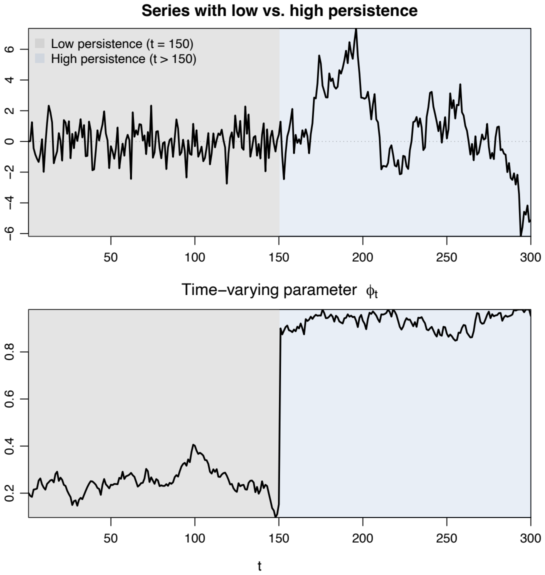
Example: DFM with Simulated Data



Example: Time-Varying Parameters (TVP-AR)

- Observation equation: $y_t = \phi_t \, y_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_y^2)$.
- State equation: $\phi_t = \phi_{t-1} + \nu_t, \quad \nu_t \sim N(0, \sigma_\phi^2)$.
- Write as SSM with state $\alpha_t = \phi_t$, observation matrix $H_t = y_{t-1}$ (time-varying H).

Example: TVP-AR with Simulated Data



Bayesian Estimation of State-Space Models

- Unknowns: parameters
$$\Theta = \{\boldsymbol{H}, \boldsymbol{F}, \boldsymbol{R}, \boldsymbol{Q}, (\text{optionally } \boldsymbol{A}, \boldsymbol{B})\},$$
  and the latent states $\boldsymbol{\alpha}_{1:T}$.
- Goal: posterior distribution
$$p(\Theta, \boldsymbol{\alpha}_{1:T} \mid \boldsymbol{Y}).$$
- Use Gibbs sampling:
  - Draw $\boldsymbol{\alpha}_{1:T} \mid \Theta, \boldsymbol{Y}$ via FFBS (forward-filtering, backward-sampling).
  - Draw $\Theta \mid \boldsymbol{\alpha}_{1:T}, \boldsymbol{Y}$ from conjugate blocks (e.g., Normal for coefficients, inverse-Wishart for covariances) or via Metropolis–Hastings.

Forward-Filtering, Backward-Sampling (FFBS)

- Forward (Kalman filter): compute $\boldsymbol{\alpha}_{t|t-1}, \boldsymbol{P}_{t|t-1}$ and $\boldsymbol{\alpha}_{t|t}, \boldsymbol{P}_{t|t}$ for $t = 1, \dots, T$.
- Backward (sampling): draw $\boldsymbol{\alpha}_T \sim \mathcal{N}(\boldsymbol{\alpha}_{T|T}, \boldsymbol{P}_{T|T})$, then for $t = T-1, \dots, 1$

$$\boldsymbol{\alpha}_t \sim \mathcal{N}\left(\boldsymbol{\alpha}_{t|t} + \boldsymbol{J}_t(\boldsymbol{\alpha}_{t+1} - \boldsymbol{\alpha}_{t+1|t}),\ \ \boldsymbol{P}_{t|t} - \boldsymbol{J}_t \boldsymbol{P}_{t+1|t} \boldsymbol{J}_t'\right),$$

where

$$\boldsymbol{J}_t = \boldsymbol{P}_{t|t} \boldsymbol{F}' (\boldsymbol{P}_{t+1|t})^{-1}.$$

- Iterate within a Gibbs loop with parameter updates.

Summary of the Kalman Filter

Given the initial conditions $\boldsymbol{\alpha}_{0|0}$, $\boldsymbol{P}_{0|0}$ and the parameter matrices ($\boldsymbol{A}$), ($\boldsymbol{B}$), $\boldsymbol{H}$, $\boldsymbol{F}$, $\boldsymbol{R}$ and $\boldsymbol{Q}$ we iterate through the following steps for $t = 1, 2, ..., T$

Forecasting steps

$$\begin{aligned}
\boldsymbol{\alpha}_{t|t-1} &= \boldsymbol{F}\boldsymbol{\alpha}_{t-1|t-1} \\
\boldsymbol{P}_{t|t-1} &= \boldsymbol{F}\boldsymbol{P}_{t-1|t-1}\boldsymbol{F}' + \boldsymbol{Q} \\
\boldsymbol{y}_{t|t-1} &= \boldsymbol{H}\boldsymbol{\alpha}_{t|t-1} \\
\boldsymbol{S}_{t|t-1} &= \boldsymbol{H}\boldsymbol{P}_{t|t-1}\boldsymbol{H}' + \boldsymbol{R}
\end{aligned}$$

Updating steps

$$\begin{aligned}
\boldsymbol{\alpha}_{t|t} &= \boldsymbol{\alpha}_{t|t-1} + \boldsymbol{P}_{t|t-1}\boldsymbol{H}'\boldsymbol{S}_{t|t-1}^{-1}(\boldsymbol{y}_t - \boldsymbol{y}_{t|t-1}) \\
\boldsymbol{P}_{t|t} &= \boldsymbol{P}_{t|t-1} - \boldsymbol{P}_{t|t-1}\boldsymbol{H}'\boldsymbol{S}_{t|t-1}^{-1}\boldsymbol{H}\boldsymbol{P}_{t|t-1}
\end{aligned}$$